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Chances Are

By **Steven Strogatz** April 25, 2010 5:00 pm

Have you ever had that anxiety dream where you suddenly realize you have to take the final exam in some course you've never attended? For professors, it works the other way around — you dream you're giving a lecture for a class you know nothing about.

That's what it's like for me whenever I teach probability theory. It was never part of my own education, so having to lecture about it now is scary and fun, in an amusement park, thrill-house sort of way.

Perhaps the most pulse-quickenning topic of all is “conditional probability” — the probability that some event A happens, given (or “conditional” upon) the occurrence of some other event B. It's a slippery concept, easily conflated with the probability of B given A. They're not the same, but you have to concentrate to see why. For example, consider the following word problem.

Before going on vacation for a week, you ask your spacey friend to water your ailing plant. Without water, the

plant has a 90 percent chance of dying. Even with proper watering, it has a 20 percent chance of dying. And the probability that your friend will forget to water it is 30 percent. (a) What's the chance that your plant will survive the week? (b) If it's dead when you return, what's the chance that your friend forgot to water it? (c) If your friend forgot to water it, what's the chance it'll be dead when you return?

Although they sound alike, (b) and (c) are not the same. In fact, the problem tells us that the answer to (c) is 90 percent. But how do you combine all the probabilities to get the answer to (b)? Or (a)?

Naturally, the first few semesters I taught this topic, I stuck to the book, inching along, playing it safe. But gradually I began to notice something. A few of my students would avoid using "Bayes's theorem," the labyrinthine formula I was teaching them. Instead they would solve the problems by a much easier method.

What these resourceful students kept discovering, year after year, was a better way to think about conditional probability. Their way comports with human intuition instead of confounding it. The trick is to think in terms of "natural frequencies" — simple counts of events — rather than the more abstract notions of percentages, odds, or probabilities. As soon as you make this mental shift, the fog lifts.

This is the central lesson of "Calculated Risks," a fascinating book by Gerd Gigerenzer, a cognitive psychologist at the Max Planck Institute for Human Development in Berlin. In a series of studies about medical and legal issues ranging from AIDS counseling to the interpretation of DNA fingerprinting, Gigerenzer explores how people miscalculate risk and uncertainty. But rather than scold or bemoan human frailty, he tells us how to do better — how to avoid "clouded thinking" by recasting conditional probability problems in terms of natural frequencies, much as my students did.

In one study, Gigerenzer and his colleagues asked doctors in Germany and the United States to estimate the probability that a woman with a positive mammogram actually has breast cancer, even though she's in a low-risk

group: 40 to 50 years old, with no symptoms or family history of breast cancer. To make the question specific, the doctors were told to assume the following statistics — couched in terms of percentages and probabilities — about the prevalence of breast cancer among women in this cohort, and also about the mammogram’s sensitivity and rate of false positives:

The probability that one of these women has breast cancer is 0.8 percent. If a woman has breast cancer, the probability is 90 percent that she will have a positive mammogram. If a woman does not have breast cancer, the probability is 7 percent that she will still have a positive mammogram. Imagine a woman who has a positive mammogram. What is the probability that she actually has breast cancer?

Gigerenzer describes the reaction of the first doctor he tested, a department chief at a university teaching hospital with more than 30 years of professional experience:

“[He] was visibly nervous while trying to figure out what he would tell the woman. After mulling the numbers over, he finally estimated the woman’s probability of having breast cancer, given that she has a positive mammogram, to be 90 percent. Nervously, he added, ‘Oh, what nonsense. I can’t do this. You should test my daughter; she is studying medicine.’ He knew that his estimate was wrong, but he did not know how to reason better. Despite the fact that he had spent 10 minutes wringing his mind for an answer, he could not figure out how to draw a sound inference from the probabilities.”

When Gigerenzer asked 24 other German doctors the same question, their estimates whipsawed from 1 percent to 90 percent. Eight of them thought the chances were 10 percent or less, 8 more said 90 percent, and the remaining 8 guessed somewhere between 50 and 80 percent. Imagine how upsetting it would be as a patient to hear such divergent opinions.

As for the American doctors, 95 out of 100 estimated the woman’s probability of having breast cancer to be

somewhere around 75 percent.

The right answer is 9 percent.

How can it be so low? Gigerenzer's point is that the analysis becomes almost transparent if we translate the original information from percentages and probabilities into natural frequencies:

Eight out of every 1,000 women have breast cancer. Of these 8 women with breast cancer, 7 will have a positive mammogram. Of the remaining 992 women who don't have breast cancer, some 70 will still have a positive mammogram. Imagine a sample of women who have positive mammograms in screening. How many of these women actually have breast cancer?

Since a total of $7 + 70 = 77$ women have positive mammograms, and only 7 of them truly have breast cancer, the probability of having breast cancer given a positive mammogram is 7 out of 77, which is 1 in 11, or about 9 percent.

Notice two simplifications in the calculation above. First, we rounded off decimals to whole numbers. That happened in a few places, like when we said, "Of these 8 women with breast cancer, 7 will have a positive mammogram." Really we should have said 90 percent of 8 women, or 7.2 women, will have a positive mammogram. So we sacrificed a little precision for a lot of clarity.

Second, we assumed that everything happens exactly as frequently as its probability suggests. For instance, since the probability of breast cancer is 0.8 percent, exactly 8 women out of 1,000 in our hypothetical sample were assumed to have it. In reality, this wouldn't necessarily be true. Things don't have to follow their probabilities; a coin flipped 1,000 times doesn't always come up heads 500 times. But pretending that it does gives the right answer in problems like this.

Admittedly the logic is a little shaky — that's why the textbooks look down their noses at this approach,